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Oefeningenreeks 5i nr. 2.4.

$f(x) = \ln(x + \sqrt{x^2 - 1})$ op interval $[1, 3]$

$$\begin{aligned} f'(x) &= \frac{1}{\sqrt{x^2 - 1}} \\ &= \int_1^3 \sqrt{1 + \left(\frac{1}{\sqrt{x^2 - 1}}\right)^2} dx \\ &= \int_1^3 \frac{x}{\sqrt{x^2 - 1}} dx \\ &= [\sqrt{x^2 - 1}]_1^3 \end{aligned}$$

$$L = 2\sqrt{2}$$

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References